CLASS IX - SUBJECT CODE 402

Chapter 1: Introduction to IT-ITeS Industry

To Remember

Data = Facts (Raw).

Information = Processed Data.

IT helps businesses reduce cost and maximize profit.

1. Introduction

In today's digital era, ICT is a basic requirement. It is difficult to think of any event without digital devices. The ITeS sector has significantly contributed to the Indian economy.

2. Information Technology (IT)

Definition: Information Technology (IT)

IT means creating, managing, storing, and exchanging information. It includes all computer hardware and software technologies used to deal with information.

Data vs. Information

Computer takes **data** as input, processes it, and produces the result as **output** (Information).

Key Definitions

Data: Refers to raw facts or raw material which are processed to get information. *(e.g., Number of boys and girls in a class)*

Information: The conclusion drawn based on processed data. Decisions are taken based on information.

Benefits of IT in Business

Reaching more potential customers.

Streamlining operations and improving efficiency.

Reducing costs and maximizing profit.

Providing better service to customers.

3. IT enabled Services (ITeS)

Definition: ITeS

IT that enables the business by improving the quality of service is ITeS. It is also called web-enabled services or remote services.

To Remember

ITeS = Outsourcing processes enabled by IT.

BPO reduces cost while controlling quality.

India is a global hub for BPO services.

ITeS covers operations that exploit IT for improving efficiency. Career options include: Call centers, payrolls, medical billing, medical transcription, back-office operations, GIS, and HR services.

4. BPO (Business Process Outsourcing) Services

Definition: BPO

BPO means performing business operations through an outside service provider. It comes under IT services as IT optimizes business performance.

Common BPO Services

Category	Examples of Services Provided
Financial & Taxation	Financial and Accounting Services, Taxation and Insurance Services
Content & Media	E-Publishing, Web Promotion, Multimedia and Design Services
Technical & Support	Software Testing Services, Health Care Services, Legal Services
Management	Document Management Services, Content Writing

5. BPM Industry in India

The IT BPM industry has fueled India's growth by contributing to the GDP and exports. It creates employment, raises income levels, and promotes exports.

Why is BPO doing well in India?

1. BPO service providers in India invest in hi-tech hardware and software.
2. They follow strict quality checks to ensure error-free service.
3. The Government of India encourages BPOs by providing logistical support.
4. Highly developed capability to deliver exceptional quality.

To Remember

IT-BPM industry contributes heavily to India's GDP.

MNC = HQ outside India.

ISP = HQ in India.

GIC = Works ONLY for parent company.

6. Structure of the IT-BPM Industry

Organizations are categorized by: Sector served, Type of offering, Geographic spread, and Size of operations.

(a) MNCs
Have their headquarters outside India but operate in multiple locations worldwide, including India. They serve external global and domestic clients.

(b) ISPs
Organizations whose headquarters are in India but may have offices internationally. They serve both global and domestic clients.

(c) GIC
These cater **only** to the needs of their parent company and do not serve external clients. Allows keeping IT operations in-house cost-effectively.

7. IT Applications (Part 1)

Computer is a tool that provides desired information. Use of computers can be observed in all aspects of life.

> **To Remember**
>
> PC = Personal Computer.
>
> Library books use Barcodes for tracking.
>
> Offices use IT for automation & efficiency.

A. IT in Home Computing

A PC is used to work at home, do household accounts, surf the web, use e-mail, play games, and pursue hobbies. PCs with CD-ROM drives and speakers play audio/video. They are also used for online shopping and e-commerce.

B. IT in Everyday Life

We use daily products embedded with software like washing machines and microwave ovens. We store appointments, contacts, and important work details digitally.

C. IT in Library

> Modern libraries are computerized. Each book has a **barcode**. This makes it easier to keep track of books and their availability. Software is used to issue and return books. Magnetic strips are used for security.

D. IT at Workplace

Computers make office work more effective. In assembly-line industries where speed and detail are critical, **automation** is becoming very common. Internet and office applications form the basis of modern business.

8. IT Applications (Part 2 - Education)

Computers are extensively used for teaching-learning and assessment. Software is used to transmit info via pictures, audio, video, and animation.

To Remember

NCERT books are available online on ePathshala.

LMS manages school systems virtually.

WBT brings learning to homes.

(a) ICT in the Classroom

- E-learning classrooms
- Videos on experiments
- Desktop publishing of magazines
- Smart-board presentations
- Educational games
- Internet gathering

(b) Education - Anywhere Anytime

Students can access NCERT books online via the **ePathshala** website/app. Students can use WBT to contact teachers via the Internet, taking education to unreached areas.

(c) Teaching Aids & Media

Used to explain difficult subjects via animations/audio-visuals, organize lessons, and make lessons interesting.

(d) Learning Management System

Benefits of LMS

Learn lessons anytime and anywhere.

Submit queries, get replies, and comment on forums.

Participate in co-curricular activities via video.

Parents can monitor their child's progress.

9. IT Applications (Part 3)

> **To Remember**
>
> CAD/CAM are used by engineers for 3D design.
>
> E-commerce = buying/selling on internet.
>
> Communication tools: email, FTP, telnet.

A. IT in Entertainment

Internet is a major source of entertainment. People download movies, play games, chat, and use multimedia. Digital broadcasting provides interactive TV programming.

B. IT in Communication

Used to convey messages, pictures, or speeches clearly. Computers are used for **email, chatting, FTP, telnet, and video conferencing.**

C. IT in Business

Computers do payroll calculations, budgeting, sales analysis, and stock management. Business over internet is called **e-commerce**. Smart cards (credit/debit cards) have a metallic strip storing the user's PIN and account number.

D. IT in Science and Engineering

> Scientists use computers for complex calculations. CAD or CAM applications are used for drawing, simulating, and testing designs. Also used for space exploration and rocket launching.

10. IT Applications (Part 4)

A. IT in Banking

Banking is now completely online. Features include: Recurring Deposits (e-RD), Fixed Deposits (e-FD), Money transfers via NEFT and RTGS. ATMs are used for deposits, withdrawals, and balance checks.

To Remember

ATMs are used for cash deposit/withdrawal.

NEFT/RTGS transfer money online.

e-Governance initiatives like Digital India help citizens.

B. IT in Insurance

Companies keep records up to date using computer databases. Tracking policy continuation, next installments, maturity dates, and bonuses is automated. Policies can be purchased online.

C. IT in Marketing

Used for advertising products. Arts and graphics create interesting advertisements. E-commerce allows people to shop from home.

D. IT in Government and Public Service

e-Governance

Government, NGOs, and International agencies use ICT to communicate and provide services to citizens. Examples: **Digital India**, Income Tax, PAN card prep, Online electricity bills, and Electronic voting.

11. IT in Health Care

Hospital Management System maintains patient records and hospital administration. Expert systems help in early disease diagnosis.

Medical Diagnostic Machines

To Remember

ECG = Heart beats.

EEG = Brain activities.

MRI = Internal organs using magnets.

CAT = 3D Body images.

Machine	Purpose & Function
CAT Machine	Creates three-dimensional (3D) images of different body parts to help diagnose diseases.
MRI Machine	Gives digital impressions of internal organs using strong magnetic fields and radio waves.
ECG Machine	Monitors the heartbeat. Records electrical impulses produced when the heart pumps blood and shows them as a graph.
EEG Machine	Records the activities of the brain using electrical probes attached to the head. Works while awake or asleep.
Cardiac Screening	Displays the physiology and movements inside the heart. Detects thinning of veins.
Blood Sugar Tester	Analyses a blood sample to determine the blood glucose level.
BP Machine	Worn as a wrist band to measure blood pressure at rest or during physical activity.

📌 Memorable Definitions

Key concepts to master for your exams

Data

Facts or raw material which are processed to get information (e.g., Number of boys and girls in a class).

Information

The result of data processing. It is the conclusion drawn based on data, used for making decisions.

Information Technology (IT)

Creating, managing, storing, and exchanging information using computer hardware and software technologies.

IT enabled Services (ITeS)

Outsourcing of processes that can be enabled with Information Technology to improve the efficiency and quality of service of an organization.

Business Process Outsourcing (BPO)

Performing business operations through an outside service provider to reduce costs while maintaining quality.

e-Governance

The use of ICT applications by the Government and NGOs to communicate and provide various services directly to the people.

🔠 Important Full Forms

All Acronyms from Chapter 1 (Underlined in the notes)

IT: Information Technology

ICT: Information & Communication Technology

ITeS: Information Technology enabled Services

BPO: Business Process Outsourcing

BPM: Business Process Management

MNC: Multinational Company

ISP: Indian Service Provider

GIC: Global In-house Center

GDP: Gross Domestic Product

HR: Human Resource

GIS: Geographical Information System

NGO: Non-Governmental Organization

PC: Personal Computer

CD-ROM: Compact Disc Read-Only Memory

LMS: Learning Management System

WBT: Web-Based Training

NCERT: National Council of Educational Research and Training

FTP: File Transfer Protocol

CAD: Computer Aided Design

CAM: Computer Aided Manufacturing

PIN: Personal Identification Number

PAN: Permanent Account Number

ATM: Automated Teller Machine

NEFT: National Electronic Funds Transfer

RTGS: Real Time Gross Settlement

MRI: Magnetic Resonance Imaging